

Chapter 2: Prompt Engineering & AI Content Creation

Class 11: Practical / Activities (10 Minutes)

Topic: Optimize and Improve AI Prompts

Activity 1: Optimize an AI Prompt (4 Minutes)

Objective

To help students learn how to improve AI prompts by adding clarity, context, role, constraints, and output format to generate more accurate, relevant, and professional AI responses.

Materials Required

- Computer/Laptop or Tablet
 - Internet Connection
 - AI Tool (ChatGPT, Google Gemini, Microsoft Copilot, or Perplexity AI)
 - Student Prompt Optimization Worksheet
 - Notebook & Pen
 - Teacher Observation Sheet
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Instructions

1. Open any AI tool.
2. Choose one topic from the list below.
3. Write a simple (basic) prompt.
4. Observe the AI-generated response.
5. Improve the prompt by adding:
 - Role
 - Task
 - Context

- Constraints
 - Output Format
6. Generate the improved response.
 7. Compare both responses and record your observations.

Suggested Topics

- Artificial Intelligence
- Climate Change
- Renewable Energy
- Cyber Safety
- Digital Marketing
- Healthy Lifestyle
- Water Conservation
- Space Exploration
- Entrepreneurship
- Smart Education

Example

Basic Prompt

"Explain Artificial Intelligence."

Optimized Prompt

"Act as a Computer Science teacher. Explain Artificial Intelligence for Class X students using simple English. Include a definition, five key features, three real-life applications, advantages, disadvantages, and a short conclusion in bullet-point format."

Student Prompt Optimization Worksheet

Task	Completed (✓)
Topic Selected	☐
Basic Prompt Written	☐
Initial AI Response Generated	☐
Prompt Optimized	☐

Task	Completed (✓)
Improved AI Response Generated	☐
Final Prompt Completed	☐

Student Activity Checklist

Students should:

- ✓ Select a suitable topic.
 - ✓ Write a basic prompt.
 - ✓ Improve the prompt using prompt engineering techniques.
 - ✓ Compare AI responses.
 - ✓ Record observations.
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Teacher Observation Checklist

- ✓ Writes clear prompts.
 - ✓ Applies prompt optimization techniques.
 - ✓ Compares AI outputs effectively.
 - ✓ Demonstrates analytical thinking.
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Activity 2: Compare and Refine AI Responses (3 Minutes)

Objective

To help students evaluate AI-generated responses, identify weaknesses, and refine prompts to improve accuracy, clarity, and completeness.

Materials Required

- AI Tool
 - Prompt Comparison Worksheet
 - Teacher Evaluation Sheet
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Instructions

1. Compare the AI responses generated from:
 - Basic Prompt
 - Optimized Prompt
 2. Evaluate each response based on:
 - Accuracy
 - Detail
 - Organization
 - Readability
 - Relevance
 3. Write one final refined prompt.
 4. Generate the final improved response.
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Prompt Comparison Worksheet

Evaluation Criteria	Basic Prompt	Optimized Prompt
Clear Instructions	☐	☐
Better Organization	☐	☐
Relevant Information	☐	☐
Proper Format	☐	☐
High-Quality Response	☐	☐
Easy to Understand	☐	☐

Discussion Questions

1. Which prompt produced the best response?
 2. What additional information improved the AI response?
 3. Why is prompt refinement important?
 4. How can optimized prompts save time?
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Teacher Observation Checklist

- ✓ Evaluates AI responses critically.
 - ✓ Refines prompts successfully.
 - ✓ Explains improvements clearly.
 - ✓ Participates actively.
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Activity 3: Present the Optimized Prompt & Reflection (3 Minutes)

Objective

To help students present their optimized prompt, explain the improvements made, and reflect on the importance of prompt refinement in AI-assisted learning.

Materials Required

- Final Optimized Prompt
 - Reflection Worksheet
 - Teacher Observation Sheet
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Instructions

1. Present your optimized prompt in **1 minute**.
 2. Explain:
 - Original prompt
 - Changes made
 - Final prompt
 - Improvement in AI response
 3. Complete the reflection worksheet.
 4. Share one lesson learned about prompt optimization.
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Reflection Worksheet

Answer the following questions:

1. What topic did you choose?

2. Write your final optimized prompt.

3. What improvements did you make to your prompt?

4. How did the AI response improve after optimization?

5. Why is prompt optimization important in AI?

Self-Assessment Checklist

Skill	Yes Needs Practice	
I wrote a basic AI prompt.	☐	☐
I optimized the prompt successfully.	☐	☐
I compared AI-generated responses.	☐	☐
I improved the quality of the AI output.	☐	☐
I understand prompt refinement techniques.	☐	☐

Teacher Observation Checklist

- ☒ Explains prompt optimization clearly.
- ☒ Demonstrates effective prompt refinement.

- ✓ Uses AI responsibly.
 - ✓ Participates confidently.
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Mini Practical Assessment Rubric

Assessment Area	Marks
Basic Prompt Writing	20
Prompt Optimization & Refinement	20
AI Response Evaluation	20
Comparison & Improvement	20
Presentation, Reflection & Participation	20
Total	100 Marks

Teacher Observation Checklist

- ✓ Writes basic and optimized AI prompts using appropriate prompt engineering techniques.
 - ✓ Evaluates and compares AI-generated responses critically.
 - ✓ Refines prompts to improve response quality, accuracy, and relevance.
 - ✓ Demonstrates logical thinking, responsible AI usage, and effective communication.
 - ✓ Participates actively and completes all practical tasks within the allotted time.
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Practical Time Distribution

Activity	Duration
Optimize an AI Prompt	4 Minutes
Compare & Refine AI Responses	3 Minutes
Present Optimized Prompt & Reflection	3 Minutes
Total Practical Time	10 Minutes

Expected Learning Outcomes

After completing these practical activities, students will be able to:

- Write and optimize AI prompts using advanced prompt engineering techniques.
- Apply **role, task, context, constraints, and output format** to improve AI-generated responses.
- Compare basic and optimized prompts to understand their impact on AI output quality.
- Evaluate AI responses for **accuracy, clarity, relevance, organization, and completeness**.
- Refine prompts through iterative improvement to generate high-quality academic and professional content.
- Demonstrate responsible and effective use of AI for **research, presentations, reports, business communication, and creative problem-solving**.